



Chapter 13 — AI-Generated Scenario Simulation & Animated Playback



Feature summary: The operator describes a situation in natural language → the AI generates a valid scenario → the engine simulates the entire situation headlessly at high speed → the platform plays it back as a smooth, narrated animation on the live map. Working name: **"What-If Theater"**.

13.1 Concept & Motivation

Today, scenarios are authored by hand in YAML (Chapter 7). This chapter adds an AI authoring and cinematic playback layer on top:

"A flood hits Sector B while two teams are on patrol and the medical depot is low on supplies."

Within seconds, the operator watches the full situation unfold as an animation: units moving, the flood spreading, incidents firing, resources draining — accompanied by an AI-written narration of the run.

Why it matters

- **Training:** instructors spin up novel, repeatable exercises on demand — no YAML skills required.
- **Planning:** instant what-if analysis; compare two variants of the same situation side by side.
- **Showcase:** this is the platform's crown-jewel demo moment — one sentence in, an animated crisis out.

13.2 The Golden Rule — Division of Responsibility



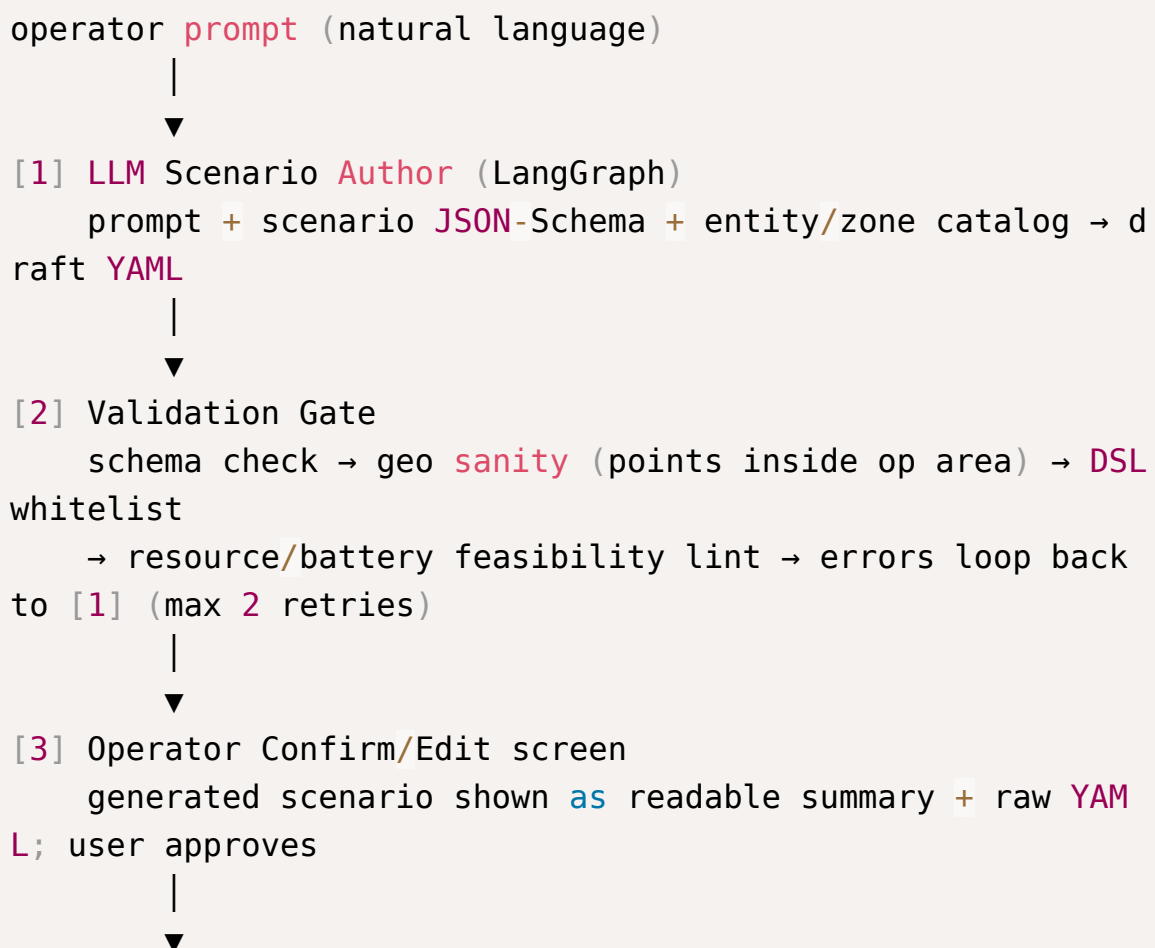
The AI authors the situation. The engine computes what happens.

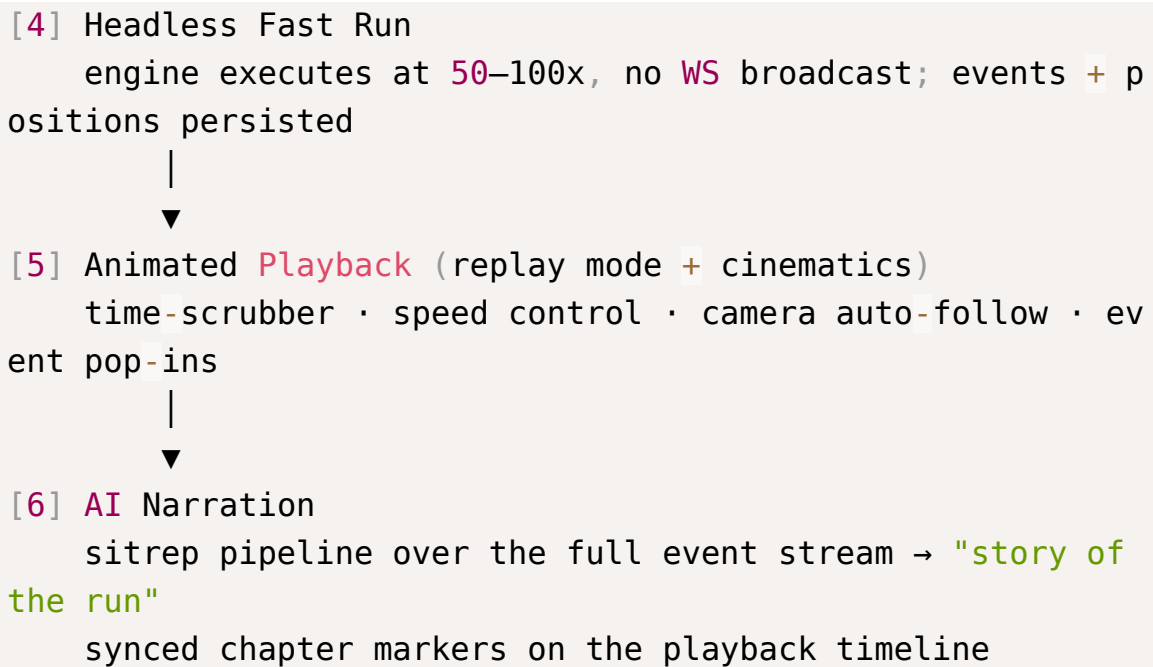
The LLM only ever generates a scenario definition (entities, timeline, rules). It never decides outcomes, positions, or physics. This preserves determinism, replayability, and credibility — the AI writes the script; the engine performs it.

Consequences of this rule:

- Same generated scenario + same seed = identical animation, every time (golden-run guarantees still hold).
- AI output is fully validated before execution — hallucination risk is reduced to “scenario rejected by schema,” never “impossible physics on screen.”
- All existing subsystems (movement, weather, logistics, vitals, events, replay) are reused unchanged.

13.3 Pipeline Architecture





13.4 Functional Requirements

- FR-13.1: Natural-language prompt box ("Simulate a situation...") available to planner/admin roles.
- FR-13.2: LLM generates scenario YAML against the Chapter 7 schema, using only whitelisted DSL actions/conditions and entities/zones from the scenario catalog.
- FR-13.3: Every generated scenario passes the Validation Gate before it can run; failures show human-readable reasons.
- FR-13.4: **Mandatory confirm/edit step:** the user sees a plain-language summary (entities, timeline, twists) and can edit YAML inline before launch.
- FR-13.5: Headless simulation completes a 3-sim-hour scenario in under 60 seconds of wall time (no WebSocket fan-out during fast run).
- FR-13.6: Playback controls: play/pause, 0.5–8x speed, time scrubber, skip-to-next-event, camera auto-follow of the highest-severity active entity, free camera.
- FR-13.7: Animated layers reuse replay rendering: interpolated unit tracks, drifting weather cells, incident pop-ins with severity styling, resource-level mini-bars.

- FR-13.8: AI narration: a generated run-story with timestamped chapter markers on the playback timeline; clicking a marker seeks the animation.
- FR-13.9: Side-by-side comparison mode: two runs (e.g., same prompt, different variable) play in synchronized split view.
- FR-13.10: Generated scenarios are saved to the scenario library, tagged `ai-generated`, with the original prompt stored for provenance.
- FR-13.11: Every AI-generated scenario and narration carries the standard “AI-generated — verify before acting” label.

13.5 LLM Scenario Author — Design Notes

Prompt strategy

- System prompt embeds: the scenario JSON-Schema, the whitelisted DSL function list, the operational-area bounds, and 2 few-shot examples of prompt → YAML.
- Generation uses structured/constrained output (YAML block only); temperature kept low (≤ 0.3) for schema fidelity.
- The model is asked to include 1–2 “dramatic beats” (a twist mid-timeline) unless the user's prompt is fully prescriptive — this makes animations engaging rather than flat.

Validation Gate checks (hard failures)

Check	Example rejection
JSON-Schema conformance	unknown key <code>wether:</code>
Geometry sanity	entity start point outside operational area
DSL whitelist	condition calls non-existent function
Feasibility lint	drone mission exceeds battery by construction
Scale limits	> 200 entities or > 500 timeline items

Failure loop: validation errors are fed back to the LLM verbatim for up to 2 repair attempts; after that, the user sees the errors and can fix manually. This turns hallucination into a visible, recoverable workflow instead of a runtime bug.

13.6 Animated Playback — Design Notes

- Built entirely on the Phase-2 replay slider; “animation” = replay + polish:
 - **Client-side interpolation** between stored positions (already required for live mode) gives smooth motion at any playback speed.
 - **Camera director**: simple rule-based auto-follow — track the highest-severity open incident's nearest responding unit; cut on new critical events; ease-to-fit for multi-entity moments.
 - **Event pop-ins**: severity-styled markers with brief toast callouts, throttled to avoid clutter at high speed.
- Optional export: capture playback to a shareable video (browser MediaRecorder) for reports and portfolio reels.

13.7 Acceptance Criteria

1. Prompt “flood in Sector B during patrols, low medical stock” produces a schema-valid scenario on first or second attempt $\geq 90\%$ of trials.
2. Generated scenario → confirm → fast run → playback completes end-to-end in under 2 minutes wall time.
3. Replaying the same generated scenario with the same seed produces a pixel-equivalent animation (determinism preserved).
4. Narration chapter markers seek the animation correctly; every narration claim maps to real events (grounding gate from Chapter 8 reused).
5. An invalid AI output (forced via test prompt) is rejected with readable errors and never reaches the engine.

13.8 Roadmap Placement & Effort

Slot	Work	Depends on
Phase 3.5 (preferred) or early Phase 6	Sprints A: LLM author + validation gate + confirm UI; B: headless fast-run mode + playback polish + narration sync	Engine (Ph. 1), replay slider (Ph. 2), AI service & sitrep pipeline (Ph. 3)

Estimated effort: **2 sprints (~3–4 weeks)** — low incremental cost because every heavy component already exists. Comparison mode (FR-13.9) and video export are stretch goals, cut first under time pressure (add to the §9.10 cut-list between items 2 and 3).

13.9 Risks Specific to This Feature

Risk	Impact	Mitigation
LLM produces boring/degenerate scenarios	Medium	Few-shot examples with dramatic beats; "regenerate with more intensity" button
Users mistake generated what-ifs for validated plans	High	Persistent ai-generated badge on scenario, playback, and narration; provenance (original prompt) always visible
Fast-run of heavy scenarios blocks the engine host	Medium	Headless runs execute as background jobs (Celery) with concurrency limit 1–2
Scope creep into a full cinematic editor	Medium	Camera director stays rule-based v1; no keyframing, no manual camera tools